

How Incentives Shape Human-AI Co-Creation

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Abstract

Generative AI is increasingly used to augment human productivity, but early evidence suggests that AI tools reduce the diversity of generated content, spurring concerns that our collective creative output may eventually homogenize as well. However, we argue that this emerging consensus overlooks how people realistically interact with AI and how incentives shape that interaction. In this work, we conduct a randomized controlled trial exploring how humans iteratively use AI under varying incentives for quality and originality. We find that although access to AI initially makes outputs more similar, the co-creation process helps counteract this effect: as people revise, edit, and adapt AI suggestions, their own contributions substantially restore diversity in the final outputs. Crucially, we demonstrate that incentives for originality serve as a powerful lever. When participants are rewarded for being unique, they continue to use AI but strategically rely less on its suggestions verbatim. These findings show that the impact of generative AI on creative output is not a fixed property of the technology itself, but is instead shaped by the behavioral strategies and reward structures of its users.

1 Introduction

People increasingly rely on generative AI to accomplish various tasks such as writing, coding, tutoring, and more [4]. While AI assistance can boost productivity and improve performance [22, 6, 8], there are also growing concerns that AI can generate incredibly homogeneous outputs [29, 40, 38, 21]. Recent evidence has also found that when humans use AI to brainstorm ideas [2, 7] or write using AI text completion [23, 1], their outcomes can be similarly homogenized—even when humans retain final control.

And yet, despite this overwhelming initial evidence, the long-term implications of AI homogenization remain unclear. A simple interpretation is that as AI adoption grows in any application, the collective homogeneity of content in that domain will grow as well. This logic is certainly echoed in popular discussions of “AI slop” polluting the Internet [26, 35]. After all, if AI systems offer similar suggestions across users, then widespread reliance on such systems will also lead to homogeneous end products.

We argue that this simple interpretation is flawed, as it overlooks critical features of real-world AI deployments that mediate their effects on homogenization. In particular, we highlight two shortcomings of existing findings—one conceptual and one methodological—that limit the extent to which we can hope to extrapolate to the real world.

At a conceptual level, **incentives matter**. In controlled experiments, participants without proper incentives may simply accept the first AI-generated suggestion because they want to complete the task as fast as possible. This is not necessarily a good proxy for real-world, high-stakes settings. Even if a startup founder seeking business ideas uses AI to brainstorm, we might intuitively expect them to bring their own ideas and discretion to the process instead of relying on AI-generated suggestions. Or, even if prospective students use AI to help write their college admissions essays, we might similarly expect them to invest substantial effort in tailoring their narratives to stand out within a highly competitive applicant pool. With stronger incentives for differentiation, human judgment might mitigate or overcome the homogenizing force of AI [25].

Methodologically, much of the evidence on AI homogenization comes from experimental setups with limited **interactivity** between humans and AI. For example, most studies on homogenization only allow humans to use AI to brainstorm ideas [2, 7] or for text completion [23, 1]. While these studies provide

invaluable initial evidence, real-world AI use is complex and interactive. A user may go back and forth with an AI model to brainstorm, draft, and refine an output. In other words, people **co-create** with AI, and how they co-create depends entirely on the context of production. A study participant may be willing to accept the first passable AI-generated output, whereas a discerning writer might be far more nuanced in their AI usage, soliciting feedback or targeted edits. Again, incentives are critical here. Without creating incentive structures that mirror real-world conditions, we should expect observed behavior to deviate systematically from real-world usage.

The present work: The impact of incentives on co-creation. Our work seeks to address both incentives and interactivity via a carefully designed randomized control trial for a creative writing task. We randomize participants’ access to AI, but unlike prior work [e.g., 7, 2, 23, 1], we simulate a more realistic interactive process with the AI tool: participants are free to chat with a general-purpose AI assistant however they see fit (e.g., to draft, edit, or consult). This setup also allows us to obtain rich, longitudinal data of participants’ strategies and writing trajectories throughout the session. Beyond this, we further randomize the *incentives* participants experience: half are incentivized to produce high-quality work, while the other half are incentivized to produce highly original work. While neither condition exactly matches real-world conditions, our primary insight is that the *difference* between these two incentive structures yields a *directionally correct* perspective on how market forces mediate human-AI co-creation. Our experimental setup thus enables us to ask key questions about the impacts of incentives. Do market-like incentives lead to more differentiation among AI users? And if so, how do humans change their usage of AI to produce more differentiated outputs?

Results. Our study reveals the following findings:

- **Human editing/co-creation substantially increases diversity.** When AI is used to generate full drafts, the resulting stories are strikingly similar to one another, often sharing stylistic patterns (e.g., starting with “I woke up...”). However, participants rarely accept these drafts as-is. Instead, they edit, revise, and re-prompt the AI with more specific preferences, producing final stories that are noticeably more diverse and much closer to the distribution of human-only writing. This suggests that human involvement produces a natural force away from the generic “AI voice”.
- **Incentives increase diversity with AI usage.** When participants using AI are explicitly incentivized to be unique, they achieve higher diversity than those without such incentives. However, they do not achieve the same amount of diversity as humans. We interpret this finding as *directional*: incentives increase diversity, but the size of the effect likely scales with stronger (or weaker) incentives.
- **Increased diversity through incentives comes from lower adoption on AI, even as AI usage increases.** Participants anchor their final stories less on AI suggestions when we incentivize originality. However, conditional on adoption, those incentivized to be original spend significantly more time prompting without an improvement in diversity.
- **Participants with greater AI experience tend to rely more heavily on AI suggestions,** resulting in more homogeneous stories. Despite their familiarity with these tools, they do not appear to adopt more effective or novel strategies for achieving originality.

Our results highlight that diversity emerges from the strategies people adopt when co-creating with AI under particular incentives. Participants actively probe, edit, and sometimes reject AI suggestions, but early exposure to homogeneous drafts still shapes downstream choices through anchoring. Incentives meaningfully shift these strategies: when novelty is rewarded, users work harder to steer the AI and rely less on its default outputs, partially restoring diversity.

More broadly, these findings suggest that human-AI interaction is best viewed as a strategic, adaptive process, where outcomes reflect not just model capabilities, but how users allocate effort, attention, and trust in response to the task environment. Designing AI systems—and incentives around them—thus plays a central role in determining whether co-created artifacts reflect the values we care about.

2 Related Work

Co-creation and Diversity. Our work is closely related to a growing literature on homogenization in generative AI systems. Prior work has shown that AI models, when prompted independently, tend to produce highly similar outputs [40, 38]. Beyond studying AI-generated text in isolation, an increasing body of research examines settings in which humans and AI co-create to varying degrees. For example, Doshi and Hauser [7] show that using AI for creative ideation leads to more homogeneous writing, while Agarwal et al. [1] and Padmakumar and He [23] find similar effects when AI is used for text completion. Together, these results suggest that homogenization arises not only from fully AI-generated content, but also from human exposure to AI suggestions. Our work builds on this line of research by studying homogenization in a more natural co-creation setting that allows for iterative prompting, editing, and selective adoption of AI outputs. We also explicitly manipulate incentives for diversity, allowing us to examine how users strategically adjust their interaction with AI in response to the task environment. Most closely related is Kosmyrna et al. [16], which also studies a relatively unrestricted co-creation setting. However, they focus on studying brain function and cognitive debt rather than, as we do, on how incentives shape user strategies and the diversity of outputs.

AI assistance, creativity, and reliance. A large body of work studies how AI assistance shapes human creativity. Much of the cognitive science and human-computer interaction literature advocates using generative AI to *enhance* creative processes rather than to replace human effort [42, 10, 5]. Creativity in these settings is often operationalized through divergent thinking tasks, where success is measured by the ability to generate a wide range of novel ideas [28], see e.g., Torrance tests [33]. Empirically, recent studies paint a mixed picture. Sun et al. [32] conduct a field experiment and find that AI assistance is most beneficial for workers who already engage deeply with their tasks. Other work suggests that AI alone or human-AI teams can yield short-term gains in novelty or productivity, but may undermine independent creative performance once assistance is removed [17, 13, 20, 31]. Our work complements these findings by adopting a broader notion of creativity that extends beyond ideation to include writing style, narrative structure, and thematic exploration in creative writing.

A recurring concern in this literature is the risk of overreliance on AI to achieve tasks, whether that be creative or otherwise. While human-AI collaboration can improve immediate performance, recent evidence suggests it may reduce intrinsic motivation or engagement when the tool is no longer available [41]. Related work also documents forms of cognitive offloading, in which users defer critical thinking or decision-making to AI systems [16, 36, 12]. Although our study does not examine long-term effects of AI use, we observe clear short-term costs of overreliance: participants who heavily anchor on AI-generated drafts produce less diverse stories.

Incentives in human-AI collaboration. Recent work has increasingly emphasized the role of incentives in human-AI decision-making. A central critique is that many studies overlook how strongly incentive structures shape participant behavior and, consequently, study outcomes [15]. Most related to our work is Holstein et al. [11], which shows that altering incentives can substantially reduce overreliance on AI in decision-making tasks. To our knowledge, our study is the first to explicitly manipulate incentive structures in a human-AI *co-creation* setting involving generative AI tools.

3 Experiment Design and Data

We run a pre-registered, between-subjects experiment in which participants are assigned via block randomization to one of four conditions in a 2×2 factorial design, which we elaborate below. Participants complete a short story writing task. We collect both the final outputs and detailed interaction logs (snapshots of text every 5 seconds as well as the complete transcripts of their AI conversations). The design and analysis plan were pre-registered on OSF.¹ Note that some of the analyses in this paper are exploratory, and we will flag them when relevant.

¹https://osf.io/3u8tq/overview?view_only=e1a8044dd5044bc59dd0a7d8f3fca3c2

3.1 Task and Recruitment

Participants were instructed to write a short story of 250–350 words using a browser-based rich-text editor. After revealing the task and instructions (Figures 7 and 8), participants are given at most 5 minutes to brainstorm and outline their short story. This stage gives participants the time to scaffold their ideas so that they will start writing with a concrete plan in mind (see Figure 9). After the brainstorming phase, they are taken to a writing window, where they have at most 20 minutes to write their story (see Figure 10). If a participant’s story is not within the desired word range, they are given an additional 5 minutes to adhere to the word limit, after which their story will be automatically submitted (and accepted). We also impose minimum times to ensure participants are faithfully exerting effort into the task: 1 minute for brainstorming and 5 minutes for writing.

We recruited a total of $n = 200$ participants from Prolific, with 50 participants per group. Participants received \$5 for completing the study, which took on average 25 minutes. The brainstorming phase took on average 4.5 minutes, while the writing phase took on average 19.6 minutes.

Ensuring attention in task. We incorporated extensive checks to ensure the fidelity of the online experiment. For example, the interface can only be done in full screen, without the ability to switch tabs or copy paste from outside the current window. Additionally, we implemented inactivity checks by tracking mouse and keyboard movements; after six 30-second periods of inactivity as well as numerous warnings, we discard the participant’s work. We also conducted post-hoc analyses of participants’ writing patterns and filtered out abnormally confident sessions with very little backtracking and pauses. Details on other mechanisms to ensure attention in this experiment can be found in Appendix A.2.

3.2 Treatment Groups

3.2.1 Factor 1: AI Usage

In the **SELF** conditions, participants wrote entirely on their own. In the **AI** conditions, participants had access to an integrated writing assistant designed to mimic general-purpose LLMs (see Figure 10). Participants are required to use the AI tool at least once, though they are not obligated to use any of the output nor interact with the tool further. However, we emphasized that participants are ultimately responsible for the text they submit. Note that we did not provide the AI tool during the brainstorming phase because we wanted to isolate the effect of AI assistance on writing style and execution more broadly; there already exists numerous studies investigating the impact of AI on creative ideation [7, 17, 20]. Those in the **AI** condition on average spent 18.6 minutes writing, compared to those in **SELF** who took on average 20.6 minutes. Both groups spent the same amount of time brainstorming.

3.2.2 Factor 2: Task Incentives

Apart from randomizing AI assistance, we also randomize participants into two conditions: a condition incentivizing **originality (O)** and another rewarding **technical quality (T)**. In **O**, participants are told that their bonus is a function of originality and uniqueness; in **T**, participants’ bonuses are based on organization and technique, see Figure 8. We emphasized in the **T** condition that their writing is not a function of their creativity. Participants in **O** spent marginally more time brainstorming and writing, with an average total time spent of 25.2 minutes compared to 24.3 minutes for **T**.

Originality (O)	Technical Quality (T)
“You are participating in a short story competition. There are thousands of submissions, so your goal is to stand out as much as possible. Find your voice and be as creative as possible! The short story should be around 250–350 words.” <i>Meta instruction:</i> “Your bonus will be determined based on originality and uniqueness.”	“You are starting an Intro to Writing class. Your first assignment is to create a 250–350 word short story. Your goal is to get an A by submitting a high-quality piece of work.” <i>Meta instruction:</i> “Your bonus will be determined based on the grade you receive.” <i>Grading is based on a rubric, see Figure 8.</i>

Table 1: Instructions given to the originality (**O**) and technical quality (**T**) conditions.

3.3 Datasets and Preprocessing

We take snapshots of the plain text editor every 5 seconds of activity, resulting in an incredibly rich temporal dataset of each participant’s progress throughout the task. We also save the entire conversation transcript for participants who had access to an AI tool. This dataset forms the basis for the bulk of our analyses.

In addition to the temporal dataset, we also collect information about participants’ characteristics. After the writing task, participants completed a short post-survey, which asked questions about their general experience with generative AI tools, and their experience with these tools for writing tasks specifically (both on a 5-point Likert scale from “Not at all familiar” to “Extremely familiar”). We also obtain non-identifying demographic information of participants, provided by default by Prolific. This includes self-reported ethnicity, age, sex, education level, and employment status.

Table 2 shows a balance table of participant demographics for each group. As expected, the demographics are relatively well-balanced due to the randomization design.

	AI-O	SELF-O	AI-T	SELF-T
<i>Continuous variables</i>				
Prolific Approvals	1211.38 (62.21)	1171.52 (57.80)	1193.66 (61.10)	1165.34 (61.04)
Age	40.74 (1.90)	39.86 (1.86)	41.32 (1.93)	40.04 (1.75)
AI Experience (general)	2.30 (0.14)	1.94 (0.16)	1.90 (0.15)	2.04 (0.14)
AI Experience (writing)	1.64 (0.17)	1.08 (0.16)	1.46 (0.17)	1.06 (0.15)
<i>Binary variables</i>				
Student	0.14 (0.05)	0.22 (0.06)	0.21 (0.06)	0.11 (0.05)
Sex (Male)	0.36 (0.07)	0.34 (0.07)	0.32 (0.07)	0.42 (0.07)
Residence (USA)	0.34 (0.07)	0.42 (0.07)	0.44 (0.07)	0.36 (0.07)
Primary Language (English)	0.94 (0.03)	0.96 (0.03)	0.92 (0.04)	0.96 (0.03)
<i>Ethnicity</i>				
Asian	5	5	4	2
Black	7	8	3	5
Mixed	1	3	5	1
Other	1	1	3	1
White	36	33	35	41
<i>Employment Status</i>				
NIPW	5	4	8	5
Full time	25	23	17	24
Other	8	6	8	9
Part time	11	14	9	10
Unemployed (looking)	1	3	8	2

Table 2: Balance table of participant demographics by group. For continuous and binary variables, we display the mean and standard error (in parentheses). All groups consist of $n = 50$ participants. “NIPW” stands for Not in Paid Work. AI Experience is on a Likert scale from 0 to 4, with 0 being not at all familiar and 4 being extremely familiar with AI tools.

AI Usage. Those with access to an AI tool ($n = 100$) on average take 3.57 conversation turns (3.23 standard deviation), resulting in a total of 357 queries. Most of them were **drafting** requests (55%; asking for full stories), then **editing** (20%; condensing, restructuring, partial edits), **creative consulting** (14%; asking opinions or help with story), **mechanical assistance** (8%; spelling or grammar checks), and others (2%).

The vast majority of participants (91%) asked for a full draft at least once. This does not suggest that they will use or anchor on these drafts (we quantify adoption later in Section 6), but they at least observe what the AI suggested. Given the importance of AI to draft, we manually encode all AI suggestions that resulted in a valid draft.

4 Diversity of Stories

A central question that we ask is how stories become more or less diverse when participants are given access to an AI tool, under various incentive structures. All of the analyses in this section were pre-registered except for the results in Figure 1 (right).

4.1 Methods

4.1.1 Diversity/Similarity Metrics

Diversity of natural language text is, in general, hard to measure since natural language is incredibly high-dimensional. Historically, the NLP literature has focused on measuring *lexical diversity* through n-gram or bit-level analyses [19, 30]. Recent works have started to use more advanced techniques, such as passing through some embedding models and calculating the similarity of these vectors [7, 40, 9]. Unfortunately, embedding models do not provide interpretable dimensions of natural language, and there are millions of embedding models to choose from – leading to a selection problem. One can generally view embeddings as capturing a combination of *semantic meaning* and *style*. There is currently no standard and there are many perspectives from the respective works we cited. Our view is that diversity in natural language is a latent construct; instead of taking a particular view of diversity, we pre-specify a suite of diversity metrics, each providing correlated aspects of text diversity, see Table 3.

Name	Description
Embeddings (using cosine similarity)	
all-MiniLM-L12-v2 ^a	One of the most popular lightweight embedding models
all-mpnet-base-v2 ^b	One of the most popular lightweight embedding models
Qwen3-Embedding-0.6B [44]	Lightweight (<1B) but high-performing on embedding benchmarks ^c
embeddinggemma-300m [34]	Lightweight (<1B) but high-performing on embedding benchmarks ^c
Style-Embedding [37]	Intended to capture style more than content (for author attribution)
STAR [14]	Intended to capture style in social media text
byt5 [43]	Breaks into byte-level instead of token-level data (might capture style)
N-gram/token based	
Compression ratio [30]	Size of original file ÷ size of compressed file (e.g., via gZip)
BLEU [24]	Geometric mean of n -gram precision between two texts
Rouge-L [19]	Harmonic mean of longest common subsequence between two texts
n -gram diversity score* [18, 23]	$\sum_{n=1}^N (\# \text{ unique } n\text{-grams} \div \# \text{ all } n\text{-grams})$

^a <https://huggingface.co/sentence-transformers/all-MiniLM-L12-v2>

^b <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

^c <https://huggingface.co/spaces/mteb/leaderboard>

Table 3: Table of similarity metrics used, in that higher similarity means reduced diversity. * n -gram diversity score is the only diversity measure, where a higher score means more diversity.

For each participant i in cell $g \in \{\mathbf{AI}, \mathbf{SELF}\} \times \{\mathbf{O}, \mathbf{T}\}$, let X_i be their text. Let $S(X_i, X_j) \in \mathbb{R}_+$ be a similarity function. Define the participant-level outcome as the leave-one-out average similarity within the same cell g :

$$Y_i^{(g)} := \frac{1}{n_g - 1} \sum_{\substack{j \in g \\ j \neq i}} S(X_i, X_j).$$

All four n -gram/token based measures are similarity functions (in that higher similarity means reduced diversity), except for n -gram diversity score (NGDS). For the embedding models, we will use the cosine similarity of the embedding vector as S . Formally, if f is an embedding model, then $S(X_i, X_j) := \cos(f(X_i), f(X_j))$.

4.1.2 Hypothesis Tests

Fix a similarity metric D . We construct the following hypotheses:

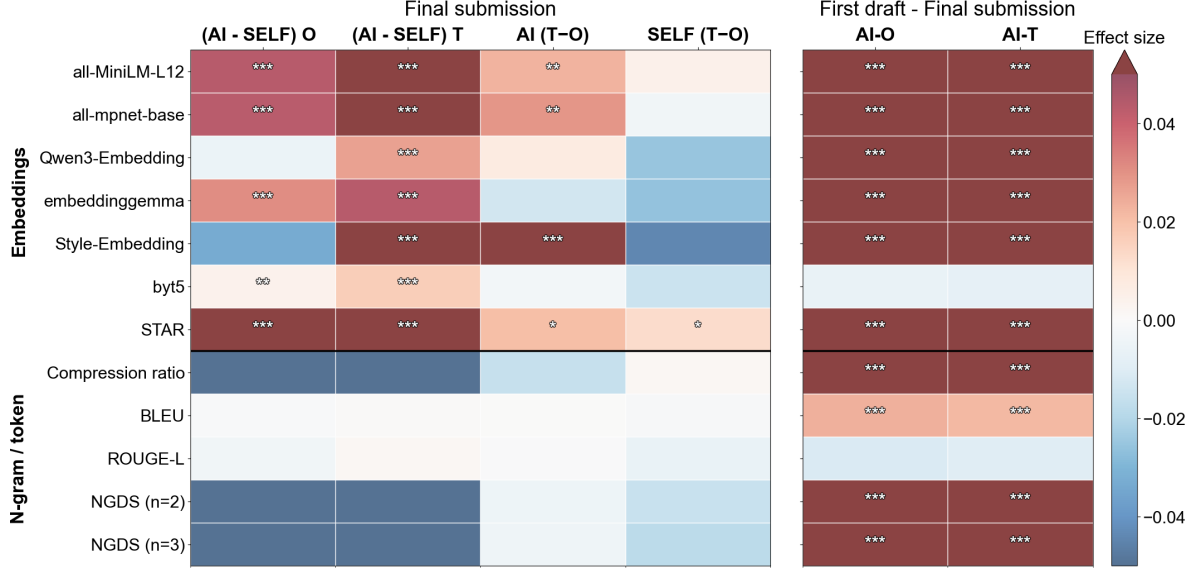


Figure 1: Effect sizes from difference in means tests (Welch t -test, one-sided) with significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, over the final submissions (left) and between the final submission and the first valid draft the AI suggested (right). A higher positive value means that the first group in the difference-in-means test is more homogeneous. All rows are similarity metrics, except for n -gram diversity score (NGDS), which is a diversity metric. As such, we flip the comparison for NGDS so that the effect sizes are directionally consistent. See Appendix Table 5 for specific effect sizes and multiple hypothesis correction.

Hypothesis 1 (Less diversity under AI).

$$H_0 : D_{\text{AI},j} - D_{\text{SELF},j} \leq 0 \quad \text{vs} \quad H_1 : D_{\text{AI},j} - D_{\text{SELF},j} > 0, \quad \forall j \in \{\text{O}, \text{T}\}$$

Hypothesis 2 (More diversity when originality is rewarded).

$$H_0 : D_{j,\text{Q}} - D_{j,\text{O}} \leq 0 \quad \text{vs} \quad H_1 : D_{j,\text{Q}} - D_{j,\text{O}} > 0, \quad \forall j \in \{\text{AI}, \text{SELF}\}$$

We construct the reverse for diversity metrics, as is the case for NGDS. From the data, we get an estimate of D_g by taking $\bar{Y}^{(g)} = \sum_{i \in g} Y_i^{(g)}$. We then conduct a two-sample difference in means test for each metric in Table 3, using Welch’s t -test because we do not assume that variances in each group are known or equal. Since these metrics are statistically correlated reflections of the same hypothesis, we do not adjust for multiple hypothesis testing. Instead, we will interpret the results as evidence about the same latent phenomenon (writing diversity), emphasizing the overall pattern across metrics rather than the significance of any single p -value. However, even with Benjamini-Hochberg adjustment for multiple hypothesis testing [3], the results remain consistently significant.

4.2 Results

Figure 1 (left) reports the results of the hypothesis tests over participants’ final submissions. In Figure 1 (right), we also display the difference in means between the first full draft that the AI produces versus the final submission. This difference gives a glimpse on how participants change and edit substantially before being satisfied with their endproduct. See Appendix Table 5 for details on specific effect sizes. Note that since we are treating each metric as a correlated view of diversity, we do not apply multiple hypothesis correction, but the results remain significant even after multiple hypothesis correction (see Table 5).

On n -gram/token based measures. First, we note that n -gram and token-based similarity metrics start off as having large effects at the first-draft stage (right subfigure), but this effect virtually disappears at final

submission (left subfigure). In contrast, embedding-based similarity measures remain statistically significant. This pattern suggests that when participants edit, they manage to diversify wording and phrasing, but deeper narrative structure and style remain more homogeneous when AI assistance is involved. We are generally more interested in characterizing higher-order features of style and narrative, and thus we will restrict our attention to the embedding-based metrics.

Hypothesis 1. We reject the null hypothesis in both incentive conditions (**O** and **T**). Across most embedding-based metrics, AI-assisted outputs are significantly more similar than human-only outputs at final submission (two leftmost columns). We also find that the first drafts provided by AI are much more similar than the AI-assisted final submissions (right subfigure), on average being 45% more similar across all metrics. This result indicates that participants often do not like the first AI draft and exert significant effort to express their own voice. Despite this additional effort, however, the AI suggestions still have a lingering homogenizing effect at final submission relative to the human-only condition.

Hypothesis 2. The evidence differs by AI access. In the **AI condition**, we reject the null (third column from left). This indicates that the incentive to be original was strong enough to induce those with AI access to make attempts to differentiate themselves. In Section 6, we will unpack the strategies participants take to achieve diversity.

In the **SELF condition**, we fail to reject the null. Stories written without AI do not show statistically significant diversity differences across incentives. One interpretation is that, absent a homogenizing tool, participants already write in their own styles, and the distinction between “quality” and “originality” incentives may be difficult to operationalize in practice. Taken together, these results suggest that incentives operate primarily through the human–AI interaction process, rather than through unaided writing alone.

5 Characterizing Writing Trajectories

In the previous section, we confirmed our hypotheses regarding the effects of AI assistance and originality incentives. We also provided an initial look at how stories evolve from the first AI draft to the final submission. In this section, we examine these trajectories in greater depth, asking *when* and *in what ways* stories shift throughout the session. From this point onward, our analyses are exploratory.

5.1 Methods

Diversity through time. We apply the same similarity metrics from Table 3, now evaluated at fixed time intervals. Taking $t = 0$ as the start of the writing session, we use snapshots of each participant’s text at 30-second intervals until the session concludes. If a participant finishes before some time $t = T$, we use their final submission for all subsequent time points. To ensure that embeddings meaningfully capture story content, we restrict attention to text containing at least 200 words. Shorter (incomplete) drafts may yield unstable or biased embedding representations.

Qualitative structure of stories. Beyond measuring diversity through time, we also ask whether different groups produce qualitatively distinct types of stories. For example, even if AI-assisted stories are less diverse, they may still occupy the same overall semantic space as **SELF** stories—or they may form a distinct cluster. To investigate this, we embed all stories and perform principal component analysis (PCA) on the embedding vectors. PCA extracts the principal axes of variation, allowing us to visualize whether stories cluster differently across conditions and through time.

5.2 Results

For clarity, we focus on results using the `all-MiniLM-L12-v2` embedding model, though the qualitative patterns hold across alternative similarity measures (see Appendix C.2).

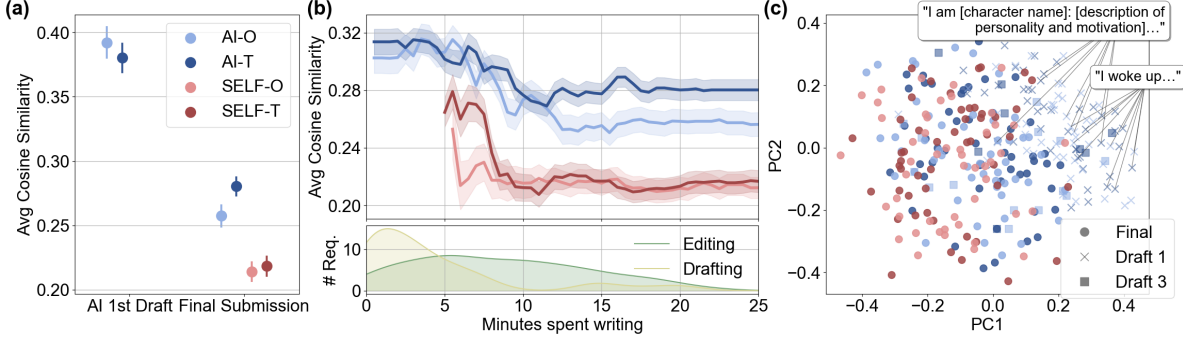


Figure 2: (a) Average cosine similarity of embeddings across randomized groups, for both the final submission and the first full draft the AI produces (when available). (b) Average cosine similarity of embeddings across randomized groups through time spent in the session. We only include text that has at least 200 words for every timestep to ensure that the embeddings capture enough information about the story. (c) First and second principal component of the embeddings across groups, for both the drafts that AI produces (when available) and the final submissions. All plots use the embedding model `all-MiniLM-L12-v2`. See Appendix C.2 for other metrics as robustness checks.

Diversity through time. Figure 2(a) reports average similarity across conditions for both the first AI draft and the final submission. These results mirror the findings from the previous section. Most strikingly, first AI drafts are highly homogeneous: in the **AI-O** condition, average similarity is 0.39, compared to 0.22 in the baseline **SELF-O** condition. But by the final submission stage, similarity in **AI-O** drops to 0.26, which is substantially closer to the SELF baseline.

Figure 2(b) traces similarity over time. Diversity increases steadily throughout the session, which is somewhat expected because participants express more of their personal style and narrative choices as the session goes on. Importantly, **AI-O** and **AI-T** collectively increase diversity after 5 minutes into the session, and begin to diverge around 12 minutes after. This divergence occurs at an inflection point where enough participants shift from asking for drafting suggestions to actively editing with or without AI (bottom subfigure). The timing suggests that incentives primarily shape how participants modify and incorporate AI outputs, rather than affecting the content of the initial AI suggestions themselves. We examine these behavioral strategies in greater detail in Section 6.

Qualitative structure of stories. We also find that apart from reduced diversity, the AI-generated stories occupy a markedly different distribution to human stories. Figure 2(c) visualizes the top two principal components of story embeddings across all stories. AI-generated drafts form a distinct cluster, separated from both AI-edited final submissions and human submissions. Qualitatively, many AI drafts share similar structures; for example, starting the story with “I woke up in...”, or introducing the main character with “I am [name], [description of character’s disposition and motivations]”. This finding indicates that there exists a recognizable AI voice that is both highly homogeneous and distinct from human writing.

A subset of participants repeatedly prompted the AI for full drafts ($n = 23$ requested at least three drafts). Across successive drafts, AI outputs move closer to the distribution of final human submissions. This shift likely reflects a combination of user steering through prompts and manual editing between queries, since we cannot rule out the possibility of participants editing themselves and then asking for full drafts given those edits. While we cannot fully disentangle these mechanisms, the pattern indicates that repeated interaction allows users to progressively shape AI outputs away from the model’s default style.

6 Differential Strategies in Using AI

Apart from characterizing the trajectories of participants’ stories, our data is uniquely positioned to answer the question of *how* participants used AI to produce their short stories. In other words, how do participants’

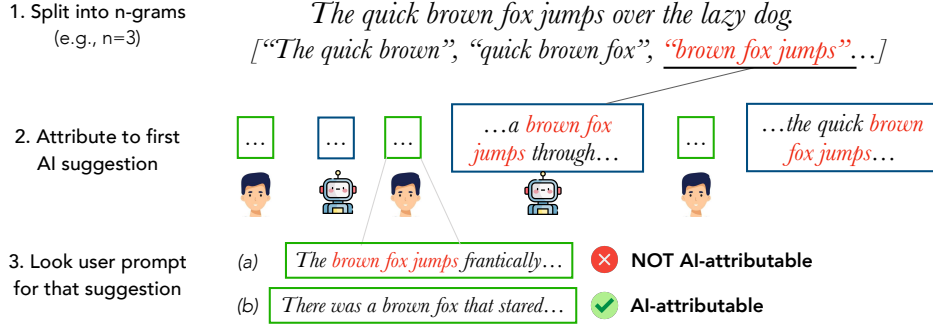


Figure 3: Example of the AI attribution procedure to construct the adoption metric.

strategies in using AI differ so that they achieve greater diversity in the **AI-O** condition? Again, we note that the following analyses are exploratory and was not pre-registered.

6.1 Methods

Measuring adoption of AI suggestions. To summarize the complex and heterogeneous trajectories participants take throughout the session, we construct a single metric that captures adoption on AI suggestions, denoted A . As we show later, this measure is highly informative of participants’ overall strategies.

Our approach is based on lexical attribution. First, we extract n -grams² for each final submission, with $n = 10$. For each n -gram, we check whether it appears in any AI-generated output within the interaction transcript. If a match is not found, the n -gram is *not AI-attributable*. If a match is found, we find the earliest AI suggestion in which the n -gram appears and check the corresponding user prompt. If the n -gram also appears in the corresponding prompt, then the n -gram is *not AI-attributable* because it was originally authored by the user. Otherwise, it is AI-attributable. See Figure 3 for a representative illustration of AI attribution. Adoption A is then defined as:

$$A = \frac{\# \text{ AI-attributable } n\text{-grams}}{\# \text{ total } n\text{-grams}}.$$

This metric is particularly useful because it accommodates the wide range of strategies participants use when interacting with AI—whether drafting entire stories, requesting partial edits, or asking for specific phrases. In Appendix B.1, we conduct multiple robustness checks: (1) we test different values of n ; (2) we also construct an embedding-based attribution measure that better accounts for semantic similarity and paraphrasing, whereas the n -gram approach cannot. All these adoption metrics are highly correlated with each other ($\rho \geq 0.94$), suggesting that our conclusions are not sensitive to the specific measure of adoption.

6.2 Results

We first examine how participants differed in their adoption of AI-generated text, and then show how these strategic differences account for the diversity patterns observed in the previous section.

Adoption as a mechanism for diversity. Figure 4(a) shows the distribution of the adoption measure A_i for participants in the **AI-O** and **AI-T** conditions. In both groups, adoption is bimodal: most participants either anchor heavily on AI suggestions or make minimal use of them. Few participants fall in between these extremes.

Relative to **AI-T**, the **AI-O** condition exhibits a clear shift toward lower adoption. More **AI-O** participants adopt a low-adoption strategy ($R_i < 0.2$), and fewer adopt a high-adoption strategy ($R_i > 0.8$). This pattern suggests that incentives to diversify encourage participants to distance their final submissions from AI-generated text, possibly because they have priors that AI outputs are relatively homogeneous or unlikely to stand out.

²A sequence of n consecutive natural language tokens, for some fixed n .

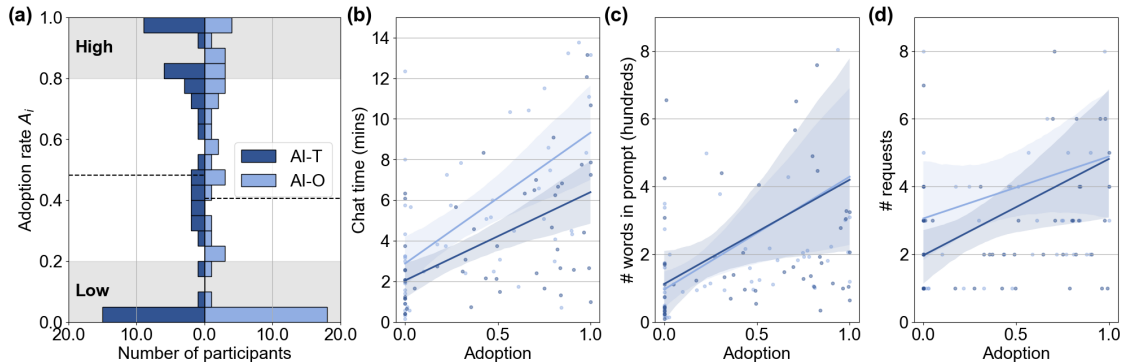


Figure 4: (a) Histogram of adoption score A_i for both the **AI-T** and **AI-O** groups. (b) Time spent writing AI prompts by adoption and group. (c) Number of words in AI prompts by adoption and group. (d) Number of AI requests by adoption and group.

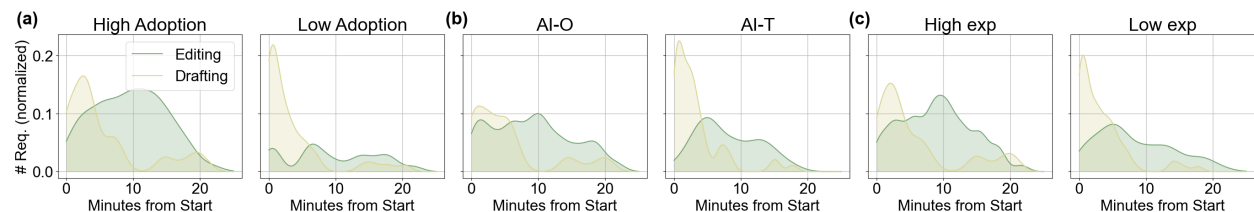


Figure 5: Distribution of editing and drafting requests normalized by number of participants in each group, over adoption levels (a), incentive group (b), and general experience with AI (c). High (low) experience is a self-reported score of ≥ 3 (< 3).

This modest shift in adoption explains the gap in diversity between **AI-O** and **AI-T**. In Figure 6, we re-estimate the difference-in-means tests after stratifying AI-assisted participants into low- and high-adoption groups. Among low-adoption participants, final submissions are nearly indistinguishable in diversity from those in the **SELF** condition; differences in similarity are small and generally not statistically significant since these participants largely author their stories independently. In contrast, high-adoption participants produce substantially more similar outputs, yielding large and statistically significant gaps relative to **SELF**.

In general, users seem to be self-aware of the strategies they take. In the post-task survey, participants were asked their strategies in using AI. Those who heavily rely on AI tend to say they keep re-prompting the AI to re-write to their liking, while those who barely rely on AI suggestions report either not liking the AI drafts or objecting to the use of AI for writing entirely. We expand on participants’ sentiments on AI usage in Section 6.3.

Low-adoption users still engage with AI. Low adoption does not imply disengagement from the AI tool. Participants in the low-adoption group still interacted with the AI multiple times, averaging just under three prompts per session (Figure 4(d)) and spending roughly three minutes composing these prompts (Figure 4(b)). These users are more likely to request full drafts than high-adoption users, but make far fewer subsequent attempts to engage with the AI (Figure 5(a)). This pattern suggests that low-adoption participants made genuine attempts to use the AI, but ultimately chose not to incorporate its suggestions—perhaps due to strong preferences over narrative direction or voice, or skepticism that the AI could further improve their story. See Section 6.3 for additional discussion of participants’ attitudes toward AI usage.

Users struggle to use AI to be unique. Figure 4(b, d) shows that participants in the **AI-O** condition exert more effort in prompting AI: across adoption levels, they prompt the AI more frequently and spend up to 50% more time interacting with it than participants in **AI-T**. However, this additional time does not translate into longer prompts (Figure 4(c)), likely because **AI-O** uses AI to edit much more than **AI-T**

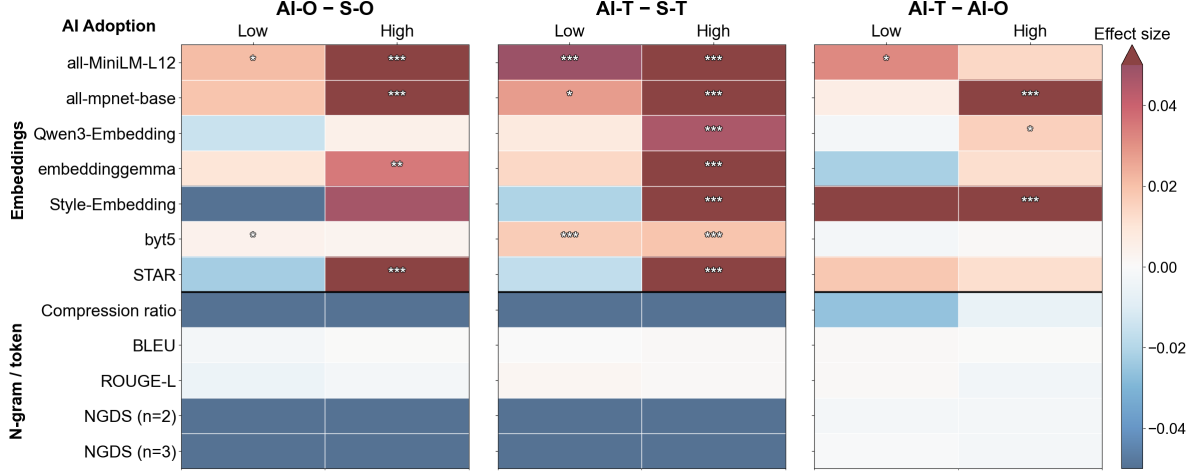


Figure 6: Effect sizes from difference in means tests (Welch t-test, one-sided) with significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, over the final submissions stratified by low and high adoption on AI among the AI-assisted group. See Appendix Table 6 for specific effect sizes.

(Figure 5(b)), and editing requests take much longer to write than drafting requests. This evidence shows that participants in **AI-O** try hard to use AI to express their unique voice via tailored edits and expressing their preferences, but this additional time and effort does not yield significant changes in diversity between **AI-O** and **AI-T**. There are various explanations for this finding: either it may be fundamentally difficult to articulate a distinctive voice through prompting alone, and most users are not highly experienced in using AI for creative writing; or, it could be that users do achieve some other goal that is not captured by our diversity metrics.

Users struggle to use AI to be unique. Figure 4(b, d) shows that participants in the **AI-O** condition exert more effort when prompting the AI. Across adoption levels, they prompt the AI more frequently and spend up to 50% more time crafting prompts than participants in **AI-T**. However, this additional effort does not translate into longer prompts (Figure 4(c)). A likely explanation is that participants in **AI-O** rely much more heavily on editing requests than those in **AI-T** (Figure 5(b)), and editing requests tend to take longer to formulate than drafting requests. Overall, this evidence suggests that participants in **AI-O** actively try to use the AI to express a more distinctive voice through targeted edits and preference-based instructions. Yet despite this additional time and effort, we observe only a slight improvement in diversity between **AI-O** and **AI-T** (see column three of Figure 1). One possible explanation is that articulating a distinctive voice through prompting alone is inherently difficult, particularly for users with limited experience using AI for creative writing. Alternatively, users may have succeeded in achieving other objectives that are not captured by our diversity metrics.

Users with more AI experience tend to anchor more on AI suggestions. Table 4 compares participant characteristics between the high- and low-adoption groups, where most demographic covariates are well balanced across the two groups. The most pronounced difference is prior experience with AI, both in general and for writing tasks specifically: participants in the high-adoption group report substantially greater AI experience than those in the low-adoption group (e.g., 2.52 vs. 1.81 for general AI experience). Unsurprisingly, participants with more AI experience tend to prompt more, spend more time interacting with the tool, and adopt similar prompting strategies to those in the high-adoption group. In particular, they are more likely to iterate with the AI by requesting partial edits throughout the session (Figure 5(c)). This pattern suggests that greater familiarity with AI is associated with increased trust in, and reliance on, AI tools. In the context of our task, however, this tendency is counterproductive because heavier anchoring on AI drafts results in lower diversity. Note, however, that we cannot rule out the possibility that these more experienced users are using AI for some other goal that our metrics do not capture, such as improving

quality. A qualitative analysis of the AI prompts show no significant differences in sophistication across experience levels.

	High adoption ($n = 27$)	Low adoption ($n = 37$)
<i>Continuous variables</i>		
Total Approvals	1221.48 (96.18)	1263.19 (60.43)
Age	39.19 (2.18)	42.16 (2.39)
AI Experience (General)	2.52 (0.21)	1.81 (0.19)
AI Experience (Writing)	2.00 (0.26)	1.14 (0.19)
<i>Binary variables</i>		
Student	0.07 (0.05)	0.16 (0.06)
Sex (Male)	0.30 (0.09)	0.41 (0.08)
Residence in USA	0.52 (0.10)	0.43 (0.08)
Primary Language (English Only)	0.93 (0.05)	0.95 (0.04)
<i>Tthnicity</i>		
Asian	14.8%	8.1%
Black	14.8%	8.1%
Mixed	7.4%	5.4%
Other	3.7%	5.4%
White	59.3%	73.0%
<i>Employment status</i>		
NIPW	11.1%	16.2%
Full time	44.4%	29.7%
Other/Expired	18.5%	18.9%
Part time	11.1%	29.7%
Unemployed (Looking)	14.8%	5.4%

Table 4: Participant demographics for high adoption and low adoption groups. For continuous and binary variables, we display the mean and standard error (in parentheses). “NIPW” stands for Not in Paid Work. AI Experience is on a Likert scale from 0 to 4, with 0 being not at all familiar and 4 being extremely familiar with AI tools.

Taken together, these results suggest that incentives to diversify do not reduce AI usage per se, but instead shape how participants integrate AI suggestions. Diversity gains arise primarily from strategic restraint in adopting AI-generated text, combined with increased effort to steer or evaluate AI outputs (though the latter effect is marginal).

6.3 User Attitudes About AI Use

We now describe users’ self-reported strategies in using AI to complete their tasks.

Participants felt agency and used AI only as a tool. Most participants treated the AI as a tool to generate an initial draft or structure, while viewing themselves as the agent who refines, edits, and enforces personal voice. Participants frequently described workflows such as “*I prompted the AI and used the text as a framework,*” “*AI laid it out nicely... I took some material from each of two drafts, then edited and altered the text to my taste,*” or “*Get a very strong baseline and then edit that manually.*” These accounts correspond to our finding that there exists substantial anchoring on AI-generated content.

At the same time, users consistently emphasized their retained control and authorship. Many explicitly referenced expressing their own writing style, describing efforts to “*make it more like my voice*” or “*adapt it to my taste.*” Notably, this language appears even among the high-adoption group. Despite relying heavily on AI drafts and often making only incremental edits, these participants report leaving the task with a sense of ownership and satisfaction. This suggests that perceived agency is preserved through repeated iteration, even when the backbone of the story originates from the AI tool. One possible explanation is that participants were given full freedom in how to interact with the AI, which may increase the subjective

sense of control. Another possibility is that high-adoption participants may have weaker priors about what would satisfy them *ex ante*; after several rounds of iteration, a partially edited output may feel sufficient in expressing their voice.

Participants were almost always unsatisfied with the AI drafts. We also found that initial AI drafts were frequently perceived as unsatisfactory (e.g., *“it gave me something that wasn’t very satisfying or quite like what I had hoped for,”* or *“[the draft] seemed quite poor”*). However, most users responded by iterating rather than abandoning the text: *“When that wasn’t very good [I would] ask it to redraft... then adjust where necessary.”* Iterative refinement appears to be the modal response to imperfect outputs.

In contrast, participants in the low-adoption group were more likely to disengage after evaluating the AI’s suggestions. Their responses often reflected disappointment with quality (*“It did not stay on point... I decided I could be better creative on my own,”* *“I tried to use the AI to assist, but it wasn’t very useful,”* *“It made it feel too computer generated.”*). Others expressed principled objections independent of quality concerns: *“I severely object to using AI in creative writing, as a novelist myself,”* or *“To ignore it almost completely. I felt it was too easy.”* These participants either abandoned the AI draft early or limited its role substantially.

Even so, low-adoption participants did not necessarily avoid the tool altogether. Many still used it for limited purposes such as inspiration (*“I used it to give me some inspiration to how to word certain settings”*) or consulting (*“I used it to assess my paragraphs and writing”*). These auxiliary uses—consulting, grammar checking, phrasing suggestions—also appear across all adoption levels.

Incentivizing originality changed how users perceived their role in co-creation. Finally, we also observe subtle but meaningful shifts in self-reported strategies between the **AI-O** and **AI-T** conditions. Participants in **AI-O** tended to self-report lower adoption on AI, frequently emphasizing that they were prioritizing creativity and personal voice (*“Add my own touches,”* *“translate it to my own voice”*). This shift in strategy suggests that participants are generally aware that AI creates a homogeneous style, and that the treatment altered how users conceptualized their role in the co-creation process.

However, these accounts do not reflect substantial differences in reported effort to prompt. Despite spending more time prompting, participants in **AI-O** rarely described that as their strategy *ex-post*. One interpretation is that users may have attempted to steer the AI toward more original outputs but, when early drafts did not meet their expectations, shifted strategies toward reducing adoption altogether. As a result, unsuccessful prompting attempts may be underrepresented in self-reports, leading participants to retrospectively frame their approach as “using AI less,” even if they initially experimented with more thoughtful prompting.

7 Discussion

Although our study focuses on diversity in creative writing, diversity is only one example of a quality we may care about in human-AI co-creation. In other domains, we might instead prioritize accuracy, safety, well-being, or learning outcomes. Our results suggest that the interaction between human behavior and AI outputs will shape whichever value is at stake.

7.1 The role of AI in co-creation

The impact of AI depends on the nature of interaction. Our findings show that the influence of generative AI on creative outputs is not determined by the model alone, but by how users interact with it. When given access to a general-purpose AI tool, people adopted a range of strategies, including requesting full drafts, asking for targeted edits, and using the model for creative consulting. Most users eventually revised and edited these outputs themselves, indicating that they retained a sense of agency over the final product. This mode of interaction differs substantially from settings where AI is used only for text completion [23, 1] or brainstorming ideas [2, 7]. Compared to these studies, we find that the iterative co-creation process reduced much of the initial homogenizing effect introduced by AI. These results suggest that whether AI

ultimately promotes or undermines certain values—such as diversity in creative outputs—depends in part on the surrounding interface and interaction design.

Incentives as a tool. At the same time, incentives remains an underexplored tool to nudge people toward certain outcomes in co-creation tasks. Our results show that incentivizing originality meaningfully altered how users positioned themselves in the co-creation process. When users were explicitly rewarded for originality, they were more likely to view themselves as authors and treat AI as a supporting aid. Stronger incentives appear to increase users’ sense of ownership and responsibility for outcomes, thereby moderating over-reliance.

Strong incentives appear naturally in many markets. For example, college admissions is one such high-stakes setting whereby many prospective students may be using AI, but may not be producing generic essays.³ In other settings, incentives could be manipulated. For example, there are concerns that junior developers are increasingly relying on AI tools, resulting in lower-quality code. In response, firms can enforce stronger incentives—such as basing promotions more on code maintainability—which may change how developers interact with AI, leveraging it as a productivity tool without compromising standards.

The broader takeaway is that incentives may be a more powerful lever than we often assume. Rather than focusing exclusively on improving model architecture or educating users about best practices, designers and organizations can shape outcomes by aligning reward structures with the values they wish to promote.

7.2 Limitations and Future Work

Long-term human-AI dynamics. Our experiment captures behavior within a single-session interaction and therefore cannot speak to how individuals’ relationships with AI evolve over time. An important open question is whether sustained use of AI might eventually help users better articulate and refine their own creative voice. While we find preliminary evidence that more experienced users tend to rely more heavily on AI without exhibiting more strategic/effective usage, this may be due to a lack of familiarity with the task. It is possible that effectively leveraging AI to achieve specific goals is itself a learned skill that develops with intentional practice. Alternatively, humans may fundamentally struggle to express their latent preferences through prompting, and increased familiarity with AI could instead reinforce over-reliance.

Technological developments may also change these dynamics. For example, AI products are becoming better at personalizing to individuals.⁴ If AI systems become capable of reliably capturing and amplifying a user’s unique voice, anchoring may no longer carry the same homogenizing consequences. Whether such hyper-personalization meaningfully preserves originality, or is even possible to begin with, remains an open question.

The evolving nature of human-AI interaction. Our experimental design was intended to approximate a realistic model of human-AI co-creation as of publication. However, AI systems and user norms are evolving rapidly, which means interaction patterns that are common today may not be as prevalent in the future. As a result, our findings should be interpreted as characterizing behavior within a particular technological and social context.

At the same time, however, the mechanisms we identify (e.g., anchoring and the role of incentives) appear consistently in behavioral studies through time. Future work should continue to examine the mechanisms through which users realistically engage with AI under varying incentives, across domains. Doing so is essential to understanding the long-term impact of generative AI tools.

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³Though, admittedly, this evidence remains to be seen in the long-run.

⁴Most large language models now have saved (long-term) memories that stores users’ preferences, see e.g., <https://help.openai.com/en/articles/8590148-memory-faq>.

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Instructions

Participant: ANON · Group: AI-DIV · Session: 5577c664-f52e-4c6a-99b8-8aef93715aff

Welcome! You'll complete a short creative writing task for a research study. Our goal is to understand how people write stories with the use of AI, in particular how it might contribute to less diverse stories. Please read the instructions carefully.

Instructions:

- Be thoughtful and authentic in your responses!
- **Stay on the session.** Once you leave the session, you will not be able to return or start a new session.
- You are provided an in-app AI tool to aid in your creative writing task; in fact you are **required to use the AI tool when prompted**. However, oftentimes the AI tool will not achieve the quality you desire. We encourage you to use the AI tool as a stepping stone and as a helper, but **ultimately you are responsible for your own story**. Please do not use external AI tools that we do not provide to you.

☐ I have read and understand the instructions.

Begin Study

Instructions

Participant: ANON · Group: SELF-CONV · Session: b5e1fc19-ca64-4c1f-a151-a96e4bcb601a

Welcome! You'll complete a short creative writing task for a research study. Our goal is to understand how people write stories without the use of AI, in particular how people brainstorm, write first drafts, and refine their stories. Please read the instructions carefully.

Instructions:

- Be thoughtful and authentic in your responses!
- **Stay on the session.** Once you leave the session, you will not be able to return or start a new session.
- **Do not use external AI tools** (e.g., Google, ChatGPT). We will be monitoring for prohibited AI usage using keystroke data, attention checks, and your final submission. **If you are caught using AI, we will return your submission.**

☐ I have read and understand the instructions.

Begin Study

Figure 7: [Top] Instructions for AI group; [Bottom] Instructions for SELF group

A Experiment Details

A.1 Interface Design

In this section, we explain the interface design in the online experiment.

A.1.1 Instruction Phase (Window 1)

Participants start with an initial instruction window, where they see different instructions depending on their randomized group, see Figure 7. If they were randomized into **SELF**, they were informed that any use of external AI tools are forbidden. If they were randomized into **AI**, they were told to only use the AI tool provided to them.

A.1.2 Prompt Phase (Window 2)

After the instruction window, they are given their writing task, see Figure 8. The only difference in the instructions is in how they are graded; those in the **DIV** group are assessed by uniqueness and originality, while those in **CONV** are assessed by the technical qualities of their writing. Since writing quality tends to be vague, we included a grading rubric to make clear that participants will be assessed by their organization (narrative flow) as well as technique (grammar, spelling, and punctuation). We emphasized that we will not be grading for creativity or uniqueness.

Prompt

You are participating in a short story competition.

There are thousands of submissions, so **your goal is to stand out** as much as possible. Find your voice and be as creative as possible! The short story should be 250-350 words.

In other words, your **bonus** will be determined based on **originality and uniqueness**.

Top % Originality	Bonus
Top 2%	\$7.00
Top 2-10%	\$4.00
Top 10-25%	\$2.00

In the next step, you will be given at most 5 minutes to brainstorm and outline your story plan. Then, you will have at most 20 minutes to write your story. You do not need to use the entire allotted time. After multiple periods of inactivity, we will flag your submission for manual review.

Please read the prompt carefully. You can proceed in 0 seconds.

Continue

Prompt

You are starting an Intro to Writing class.

Your first assignment is to create a 250-350 word short story. Your goal is to get an A by submitting a high-quality piece of work.

Grading Rubric

Your story will be evaluated on two aspects:

1. Organization (Narrative Flow)	(4 pt) Excellent: Clear beginning, middle, and end with smooth transitions (3 pt) Competent: Logical flow with minor structural issues (2 pt) Basic: Some organization but lacks coherence (1 pt) Does not meet expectations: Disorganized or unclear structure
2. Technique (Grammar, Spelling, Punctuation)	(4 pt) Excellent: No errors; demonstrates mastery (3 pt) Competent: Few minor errors that don't impede understanding (2 pt) Basic: Several errors that occasionally distract (1 pt) Does not meet expectations: Frequent errors that impede understanding
Creativity	Note: You are not graded on creativity or your voice . While creativity and uniqueness are often contributors to good pieces of writing, for now we are only asking for a technically well-written story.

Your **bonus** will be determined based on the **grade you receive**:

Point Range	Grade	Bonus
8 points	A	\$2.50
6-7 points	B	\$1.00
5 points or lower	C or lower	None

In the next step, you will be given at most 5 minutes to brainstorm and outline your story plan. Then, you will have at most 20 minutes to write your story. You do not need to use the entire allotted time. After 30 seconds of inactivity, we will flag your submission for manual review.

Please read the prompt carefully. You can proceed in 0 seconds.

Continue

Figure 8: [Top] Prompt and rubric for DIV group; [Bottom] Prompt and rubric for CONV group

A.1.3 Brainstorming phase (Window 3)

Participants are then brought to a brainstorming window, where they are given five minutes to brainstorm ideas and outline a story, see Figure 9. This window does not differ across groups. To better guide participants, we divided the outline into “Main Character”, “Setting”, “Conflict”, “Resolution”, and “Plot”. We also gave nudges that the story should be in first-person, take place over no more than one day, and should center on a decision or dilemma. While these were presented as instructions, they were intended to place guardrails on the seemingly open-ended task, since participants only had 20 minutes to write a story. Participants had to at least write 20 words in the outline, and spend at least one minute in this window to make sure that they are participating in good faith.

A.1.4 Writing phase (Window 4)

After brainstorming, participants are given 20 minutes to write their short story. We include the text from the outline in the previous window to help participants recall their writing plan. Those in **SELF** are given just a plain-text editor, while those in **AI** are given a side-by-side window where left is a plain-text editor and right is an AI conversation box. The AI assistant was `gpt5-mini` with reasoning set to medium, and we capped usage to 15 calls/prompts for each session. While they are not required to use any of the AI suggestions, participants are required to prompt the AI tool at least once.

After 20 minutes, if the text in the text editor is not within 250-350 words, participants have a 5 minute grace period, during which they are instructed to quickly make edits to adhere to the word limit. After the 5 minute grace period, their session immediately ends and the responses are automatically submitted. Only 11 submissions did not adhere to the word limit, which we still include in the final analysis because they were reasonably within the word limit.

A.1.5 Post-Survey (Window 5)

Finally, we ask three questions to all participants, and an additional question for those randomized to **AI** (see Figure 11):

1. How familiar are you with using AI for general tasks? (0-4 Likert scale)
2. How familiar are you with using AI for writing (creative or otherwise)? (0-4 Likert scale)
3. What aspects would you look out for to determine if a piece of writing is AI-generated?
4. (Unique to **AI** group) What was your strategy in using AI to complete your writing task?

A.2 Attention Checks and Safeguarding Against AI Use

Ensuring the fidelity of an online experiment has been the topic of much debate recently, especially since participants can access a generative tool externally [27, 39]. This is particularly concerning in our setting where the **SELF** group serves as an important human baseline. We took great steps to safeguard against external AI use, and in this section we discuss the strategies we employed.

Global strategies. Participants are asked to enter full screen mode. If they exit from full screen or switch windows while on full screen, they will be met with a warning as in Figure 12, which displays the number of violations so far in the session. If the participant reaches 5 violations, the session automatically ends and their submission will not be counted toward the final analysis. Whenever they are booted off the platform, the same participant ID cannot restart the session.

We also employ an **attention mechanism** throughout the session. The window has listeners to keep track of mouse and keyboard activity, though these are not saved. After 30 seconds of consecutive mouse or keyboard inactivity, participants are given a warning saying that they should stay engaged by moving their mouse or typing. In each session, they are given 6 such warnings before their session automatically ends and their submission will not be counted toward the final analysis. After the second-to-final warning, the warning persists in the window to make sure that participants know the risk of being banned from the

Step 1 • Brainstorm

Story Guidelines

To help spark ideas and make your story easier to shape, we've added a few gentle guidelines to focus your creativity:

- Write the story from a **first-person** point of view.
- The story should take place over **no more than one day**.
- The story should center on a **decision or dilemma**.

Outline your story plan below. Use as many of the boxes as you find necessary. Remember, your goal is to **get the highest grade possible!** Remember, the story should be **250-350 words**.

Quick Ideas

Jot down as many ideas for a story you have.

Outline

Main Character

Who is your main character? What are their traits? Additionally, what is your character's goal in the story? What do they want?

Who is your main character?

Setting

Where does this story take place? When does this story take place? What time period does the story occur over (1 year? 1 day? 20 minutes?)

Where and when does the story take place?

Conflict

What is the main conflict of this story? How does this conflict prevent the character from getting what they want?

What problem drives the story?

Resolution

How does the story end? Does the main character achieve their goal? Why or why not?

How is the conflict resolved?

Plot

Now, write out the events of the story. How does the story get from beginning to end? What happens? How is the resolution reached?

Summarize the main events or structure.

Figure 9: Brainstorming and outlining page.

Step 2 · Writing

Story Guidelines

To help spark ideas and make your story easier to shape, we've added a few gentle guidelines to focus your creativity:

- Write the story from a **first-person** point of view.
- The story should take place over **no more than one day**.
- The story should center on a **decision or dilemma**.

Write Your Story

0 words

Required length: 250-350 words

Write here... (250-350 words)

Note: Copy and paste is disabled

Your Brainstorm Outline

▼ Hide

Main Character

The quick brown fox jumps over the fence, and eats the bear. He is very vicious but has a soft spot for chickens.

Step 2 · Writing

Story Guidelines

To help spark ideas and make your story easier to shape, we've added a few gentle guidelines to focus your creativity:

- Write the story from a **first-person** point of view.
- The story should take place over **no more than one day**.
- The story should center on a **decision or dilemma**.

Write Your Story

0 words

Required length: 250-350 words

Write here... (250-350 words)

AI Assistant

15 queries remaining

Ask the AI for a first draft or to edit.

Type a message... (press Enter for new line)

Send

► **Tips for prompting the AI** (click to expand)

Note: Copy and paste is allowed only within the page

Figure 10: [Top] Writing interface for **SELF** group; [Bottom] Writing interface for **AI** group.

Post-Session Survey

Thank you! A few quick questions:

How familiar are you with using AI for general tasks? (i.e., what is your level of expertise in using AI?)

Select... ▾

How familiar are you with using AI for writing (creative or otherwise)?

Select... ▾

What aspects would you look out for to determine if a piece of writing is AI-generated?

Add one aspect per line. You can add or remove lines as needed. Please add at least one aspect.

Aspect 1 ✕

Aspect 2 ✕

+ Add another aspect

Briefly, what was your strategy in using AI to complete your writing task?

Describe your approach to using the AI assistant...

Please answer all questions to continue

Submit & Finish

Figure 11: Post-session survey questions.

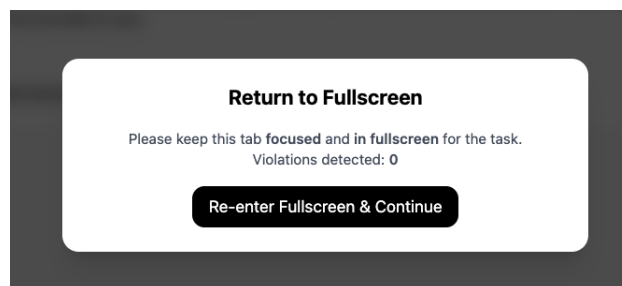


Figure 12: Warning sign instructing users to enter full screen to continue the study.

Instructions

Participant: ANON · Group: SELF-DIV · Session: 31491b95-b254-4fd7-97cb-bfa107339675

Welcome! You'll complete a short creative writing task for a research study. Our goal is to understand how people write stories without the use of AI, in particular how people brainstorm, write first drafts, and refine their stories. Please read the instructions carefully.

Instructions:

- Be thoughtful and authentic in your responses!
- **Stay on the session.** Once you leave the session, you will not be able to return or start a new session.
- **Do not use external AI tools** (e.g., Google, ChatGPT). We will be monitoring for prohibited AI usage using keystroke data, attention checks, and your final submission. **If you are caught using AI, we will return your submission.**

☒ I have read and understand the instructions.

Attention Check

Based on the instructions above, what is the policy regarding AI usage in this study?

☐ Zero tolerance - No AI usage allowed at any point

☒ AI usage is only allowed when specifically provided by the study

☐ Any AI usage is allowed throughout the study

⚠️ Attention Check Failed

You have failed to correctly identify the AI usage policy after multiple attempts. This indicates that you may not have read the instructions carefully enough.

Unfortunately, we cannot proceed with the study if participants do not understand the key requirements.

Please [click this link](#) to go back to Prolific and return the study. Alternatively, copy and paste this code: **C246RY97**.

Your **bonus** will be determined based on **the grade you receive**:

Point Range	Grade	Bonus
8 points	A	\$2.50
6-7 points	B	\$1.00
5 points or lower	C or lower	None

In the next step, you will be given at most 5 minutes to brainstorm and outline your story plan. Then, you will have at most 20 minutes to write your story. You do not need to use the entire allotted time. After 30 seconds of inactivity, we will flag your submission for manual review.

Based on the prompt, what is your bonus determined by?

☐ Originality and uniqueness
 ☐ The grade I receive

Begin Brainstorm

Figure 13: Attention checks for the Instruction phase (top) and the Prompt phase (bottom). Participants are not allowed to continue the session if they fail to answer both questions correctly.

platform. Participants are given up to 6 chances to ensure that they can take breaks (e.g., to grab water), or for slack if they are thinking for extended periods of time. If, however, there is minutes of inactivity at any given point, the session automatically ends.

Window-specific strategies. In each window, we also employ specific strategies to make sure participants are paying attention and engaged.

- Instruction phase (Window 1): Participants are asked to acknowledge that they read the instructions, and are given a quick attention check to make sure they have read the AI policy (see Figure 13, top). If they fail, their session will immediately end and they cannot restart.
- Prompt phase (Window 2): Similarly, participants are asked a question about how their writing is assessed (see Figure 13, bottom). If they fail, their session will immediately end and they cannot restart.
- Brainstorming phase (Window 3): Participants cannot copy paste anything onto the textboxes. The attention mechanism is also in place in this window.
- Writing phase (Window 4): Participants in **SELF** cannot copy paste anything onto the textbox, while those in **AI** can only copy paste within their window in case they want to integrate the AI suggestions into their own writing. The attention mechanism is also in place in this window.

Post-hoc strategies. Even after the extensive strategies employed during the session, we conducted one final analysis by leveraging the 5-second snapshots of each participant’s writing. For every participant in the **SELF** group, we analyzed the extent to which they backtracked or stopped writing—indications that they hesitate and revise—consistent with a realistic human writing process. Only one participant failed our threshold because they barely backtracked or stopped; We cannot, however, determine whether the participant is an incredibly confident writer or using external AI tools on the side, and so we decided to pay the participant but exclude their results from the study.

In all, the strategies we employed are quite conservative, and we managed to filter out 29 Prolific workers. In fact, a handful of participants had made good effort attempts but were still banned from the platform; we made sure to go through each complaint and pay each participant accordingly for their time. These mechanisms cannot, however, exclude the possibility that an extremely sophisticated participant is using an external AI tool on another device, and is writing the story with uncertainty and pause. We believe that this is an unlikely scenario to begin with, and even if they exist, are rare enough not to significantly change our overall findings.

A.3 Recruitment Details

In order to obtain high-quality submissions, we applied the following inclusion criteria: (1) Resides in the United States or United Kingdom; (2) Primary language is English or fluent in English; (3) At least 18 years old; (4) At least 500 previously completed tasks on Prolific; (5) Historical approval rate of at least 98%. We sought to recruit individuals who are experienced and have a good track record in Prolific, and can write confidently in English due to the strict time constraint.

A.4 Payment Details

Participants are paid \$5 for completing the study. On average, participants took around 25 minutes to complete, resulting in an hourly pay of around \$12. We also gave a fixed bonus of \$1 for every completed work; this is to deliver on the promise in the prompts that promised to pay bonuses based on the quality of their work. However, instead of going through each story and making a (subjective) decision on quality, we decided to give a fixed payment of \$1. Since this payment was done well after participants completed the study, it has no impact on the incentives that the prompt were intended to have. Note also that we gave partial payments for edge cases where participants failed some attention check, but already invested sufficient time into the task and were determined to have done the task in good faith.

B Methods Details

B.1 Measuring Adoption on AI-Generated Text

For each participant i , we observe a final written submission F_i and a sequence of AI-generated suggestions $\{S_{ik}\}_{k=1}^{m_i}$ produced during the interaction. These AI generations may take different forms, including full drafts, partial rewrites, or shorter textual suggestions. Our goal is to quantify the extent to which the participant’s final submission is anchored on AI-generated content.

We operationalize this notion via a single adoption measure,

$$A_i \in [0, 1],$$

which captures the fraction of the final submission that can be attributed to AI-generated text. Intuitively, A_i is high when large portions of the final submission closely resemble AI output, and near zero when the final submission is largely authored independently.

To construct A_i , we consider two complementary attribution schemes: a conservative *n-gram-based* approach that captures near-verbatim reuse, and a more permissive *embedding-based* approach that additionally captures paraphrased reuse. In both cases, the core idea is to identify which portions of the final submission can be plausibly traced to any AI generation, and to measure their total mass. We have introduced the *n-gram* based metric in the main text; here, we introduced the embedding-based approach, which we use as a robustness check for our main reliance measure.

B.1.1 N-gram based

Let $F_i = (f_1, \dots, f_{T_i})$ denote the token sequence of the final submission, where T_i is the total number of tokens. For each AI suggestion S_{ik} , we extract all contiguous *n*-grams (for a fixed *n*, e.g., $n = 10$). We then scan the final submission for maximal contiguous spans of tokens that appear verbatim in S_{ik} .

When an *n*-gram in the final submission appears in multiple AI generations, we attribute it to the *earliest* AI generation in which it appears. However, to avoid attributing content that originated from the participant rather than the AI, we further check whether this *n*-gram already appears in the corresponding user prompt that produced that AI generation. If so, the *n*-gram is not counted as AI-attributable because the participant originally generated it. Let W_i denote the number of final tokens that are AI-attributable under this criterion. The *n*-gram-based adoption measure is then defined as

$$A_i = \frac{W_i}{T_i}.$$

B.1.2 Embedding-Based Reliance (Paraphrased Reuse)

The *n*-gram approach is conservative and does not capture paraphrased reuse. We therefore define a second reliance measure based on local semantic alignment using sentence embeddings. We segment the final submission into a sequence of spans (e.g., sentences), $\{F_{iu}\}_{u=1}^U$, where each span covers a contiguous set of tokens. Each AI suggestion S_{ik} is segmented in the same way. Let $e(\cdot)$ denote a sentence embedding function.

For each final span F_{iu} , we compare it sequentially to the AI generations in the order they were produced. Specifically, we identify the *first* AI generation k for which F_{iu} exhibits sufficiently high semantic similarity to any span in A_{ik} , as measured by cosine similarity between embeddings. Formally, we consider the smallest index k such that

$$\max_v \cos(e(F_{iu}), e(S_{ikv})) \geq \tau,$$

for a fixed high threshold τ .

To avoid attributing content that originated from the participant rather than the AI, we further check whether F_{iu} is already present in the corresponding user prompt that produced generation S_{ik} , using the same embedding-based similarity criterion (and optionally additional lexical guardrails). If F_{iu} matches the prompt, it is treated as human-originated and is not counted as AI-attributable.

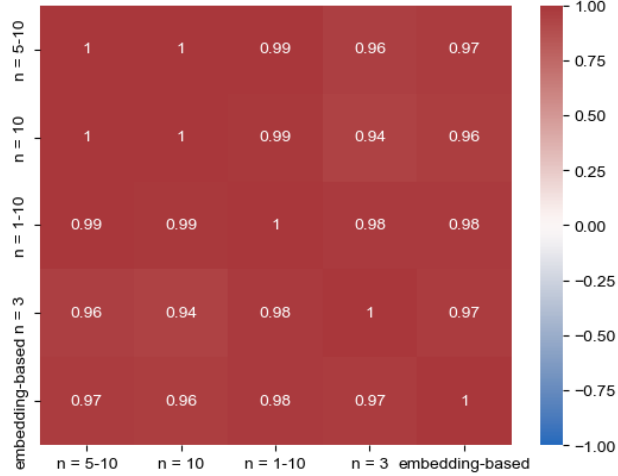


Figure 14: Correlation matrix of various adoption metrics. For example, $n = 1 - 10$ means that the adoption metric aggregates from $n = 1$ to 10.

A final span is considered AI-attributable if it satisfies the above criterion for some AI generation and does not match the corresponding user prompt. These criteria are chosen to capture close semantic reuse, including paraphrasing, while avoiding attribution to generic topical similarity or high-level advice.

Let m_u denote the token length of span u , and let

$$W_i^{\text{emb}} = \sum_{u=1}^U m_u \cdot \mathbf{1}\{F_{iu} \text{ is AI-attributable}\}$$

be the total number of final tokens attributable to AI under the embedding-based criterion. We define the embedding-based reliance measure as

$$A_i^{\text{emb}} = \frac{W_i^{\text{emb}}}{T}.$$

B.1.3 Robustness of adoption metric

In the main text, we used $n = 10$ -gram as the basis of our adoption metric. In Figure 14, we show that our choice is $n = 10$ is highly correlated ($\rho \geq 0.94$) with other choices: lower n , aggregated over multiple choices of n , or even the embedding-based metric. We therefore do not replicate our results with this metric, since the analyses will be nearly identical.

C Additional Results

C.1 Diversity of Stories

Table 5 shows the effect sizes from difference-in-means tests over the final submissions and the first valid draft the AI suggested. Red stars indicate significance after a Benjamini-Hochberg adjustment, but the results remain largely consistent.

C.2 Characterizing Writing Trajectories

In Figure 2, we plotted results for `all-MiniLM-L12-v2`. Here, we display the same plots for all pre-registered similarity (and diversity in the case of NGDS) metrics. Figure 15 shows the average similarity across randomized groups for both the final submission and the first full draft of AI. Figure 16 plots the trajectories of the similarity metric across groups through time. Finally, Figure 17 shows scatter plots of the first and

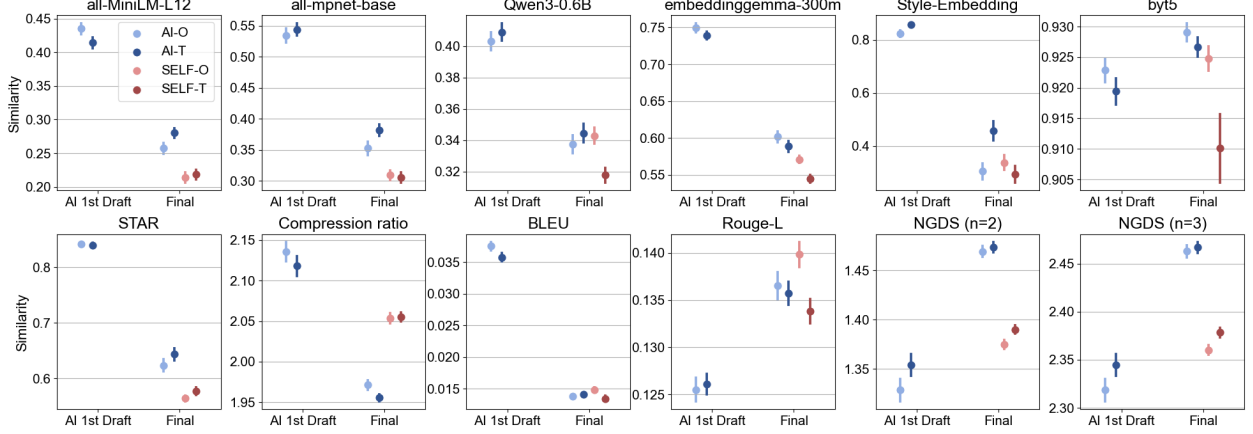


Figure 15: Average similarity (diversity for NGDS) across randomized groups, for both the final submission and the first full draft the AI produces (when available). NGDS is the diversity metric, in that a higher value means lower similarity.

second principal components (PC) from the story’s text embeddings. All the results as discussed in the main text generally hold.

C.3 Differential Strategies in Using AI

Table 6 displays the effect sizes from difference-in-means tests, where the **AI** group is stratified into high and low adoption ($A > 0.8$ and $A < 0.2$, respectively).

Comparison	AI-O – SELF-O		AI-T – SELF-T		AI (T – O)		SELF (T – O)
Version	Final	First (AI)	Final	First (AI)	Final	First (AI)	Final
Embeddings							
all-MiniLM-L12	0.043***	0.212***	0.062***	0.191***	0.023**	−0.017	0.004
all-mpnet-base	0.043***	0.227***	0.076***	0.232***	0.029**	0.001	−0.003
Qwen3-Embedding	−0.005	0.060***	0.027***	0.087***	0.007	0.001	−0.025
embeddinggemma	0.031***	0.178***	0.044***	0.189***	−0.013	−0.015	−0.026
Style-Embedding	−0.033	0.485***	0.164***	0.551***	0.153***	0.023**	−0.044
byt5	0.004**	−0.003	0.017***	0.010**	−0.002	−0.001	−0.015
star	0.059***	0.273***	0.067***	0.253***	0.020*	−0.008	0.013*
N-gram/token based							
compression ratio	−0.083	0.087***	−0.100	0.056***	−0.016	−0.029	0.002
self bleu	−0.001	0.022***	0.001	0.022***	0.000	−0.001	−0.001
rougel	−0.003	−0.016	0.002	−0.008	−0.001	0.002	−0.006
n gram diversity score 2	0.095	−0.051***	0.083	−0.030***	0.004	0.036	0.015
n gram diversity score 3	0.103	−0.048***	0.088	−0.029***	0.004	0.037	0.018

Table 5: Effect sizes from difference in means tests (Welch t-test, one-sided) over the final submissions (“Final”) and the first valid draft the AI suggested when available (“First (AI)”). Black significance stars are standard p-values, while red stars use Benjamini–Hochberg (BH) FDR-adjusted p-values: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

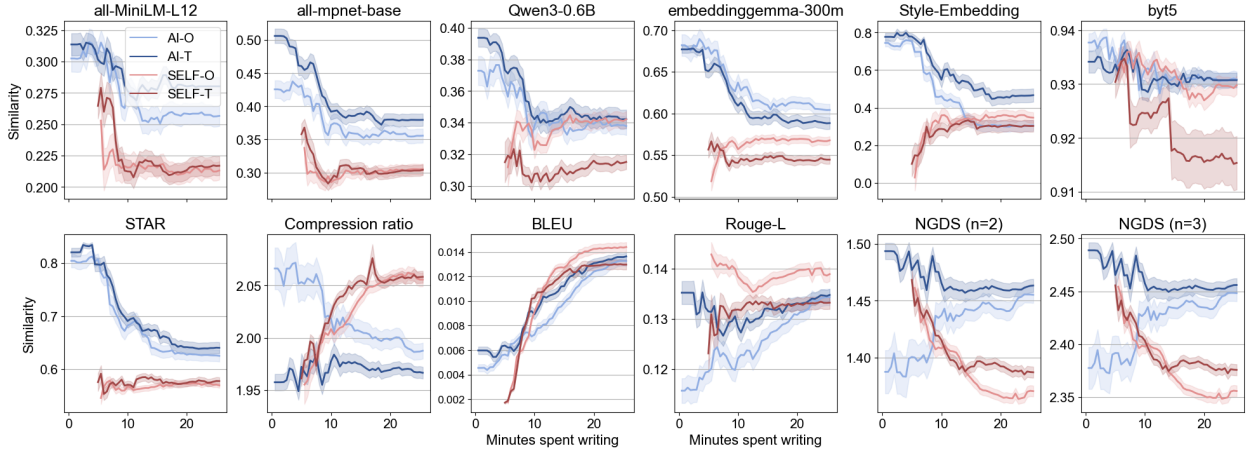


Figure 16: Average similarity (diversity for NGDS) randomized groups through time spent in the session. NGDS is the diversity metric, in that a higher value means lower similarity. We only include text that has at least 200 words for every timestep to ensure that the embeddings capture enough information about the story.

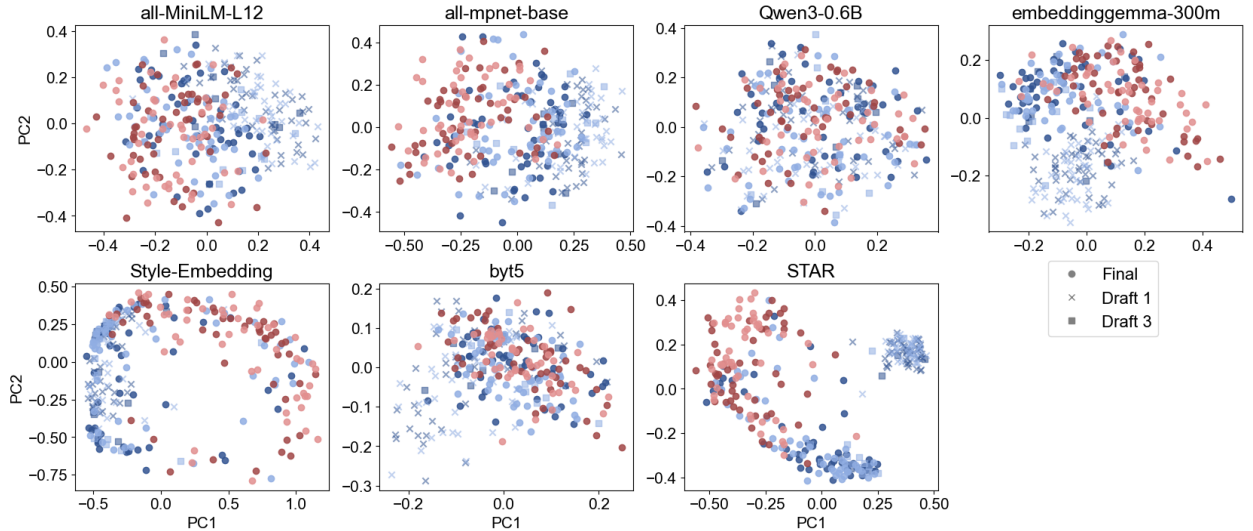


Figure 17: First and second principal component of the embeddings across groups, for both the drafts that AI produces (when available) and the final submissions.

Comparison	AI-O - SELF-O		AI-Q - SELF-Q		AI-Q - AI-O	
AI Adoption level	Low	High	Low	High	Low	High
Embeddings						
all-MiniLM-L12-v2 ↓	0.015	0.069***	0.051***	0.072***	0.041**	0.008
all-mpnet-base-v2 ↓	0.016	0.061***	0.021	0.115***	0.002	0.050***
Qwen3-Embedding-0.6B ↓	-0.016	0.009	0.004	0.043***	-0.005	0.009
embeddinggemma-300m ↓	0.007	0.036***	0.011	0.072***	-0.023	0.010
Style-Embedding ↓	-0.157	0.042	-0.045	0.261***	0.068	0.175***
byt5 ↓	0.004*	0.004	0.016***	0.016***	-0.002	-0.003
star ↓	-0.027	0.116***	-0.025	0.115***	0.015	0.011*
N-gram/token based						
Compression Ratio ↓	-0.053	-0.111	-0.079	-0.117	-0.024	-0.004
BLEU ↓	-0.002	-0.000	-0.000	0.000	0.001	-0.001
ROUGE-L ↓	-0.006	-0.003	0.002	0.000	0.002	-0.003
NGDS ($n = 2$) ↑	0.067	0.119	0.053	0.108	0.001	0.004
NGDS ($n = 3$) ↑	0.074	0.129	0.056	0.115	0.000	0.004

Table 6: Effect sizes from difference in means tests (Welch t-test, one-sided) with significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, over the final submissions stratified by low and high adoption on AI among the AI-assisted group.